

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0373

Broadford 138 kV
November 18, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0373. The Interconnection Customer proposes a 95.2 MW wind facility interconnecting at the Broadford 138 kV substation in the Virginia area. The request was received on August 30, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 218 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,176,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The IOCO – KEAN MTN AND GRASSY CRK 138 kV L is overloaded to 230.8% of its 256 MVA emergency rating for the loss of the CLINCH RIVER – CARLTON HILLS 345 kV Line. The project contributes 6.9% DFAX (6.5 MW) to this overload.

HAYSI – HOBART 138 kV L: 142.0% loading (206 MVA) following the outage of the IOCO – KEAN MTN AND GRASSY CREEK 138 kV; DFAX = 5.8%.

KEAN MTN AND GRASSY CRK 345 kV/1 kV Transformer: 139.4% loading (548 MVA) following the outage of the CLINCH RIV – CARLTON HILLS 345 kV; DFAX = 11.1%.

For the loss of CARLTON HILLS – CLINCH RIVER 345 kV, the KEAN MTN AND GRASSY CREEK 138/1kV TX loads to 139.4% of its 548 MVA rating. Project DFAX: 11.1%.

The NO TROY – LOST NATION 138 kV is overloaded to 125.7% of its 240 MVA emergency rating for the loss of the LOWMOOR – HINTON AP 345 kV line. The project contributes 14.5% DFAX (13.8 MW) to this overload.

For the loss of LOWMOOR – HINTON AP SUBSTATION 345 kV Line, the LYNN – LOST NATION 138 kV loads to 124.8% of its 239 MVA rating. Project DFAX: 14.5%.

LYNN – ROGUE 138 kV: 124.8% loading (235 MVA) following the outage of the HINTON AP – LOWMOOR 345 kV; DFAX = 14.5%.

ROGUE – TREEVANT 138 kV: 124.2% loading (233 MVA) following the outage of the HINTON AP SUBSTATION – LOWMOOR 345 kV Line; DFAX = 14.5%.

The TREZEVANT – SHULLSBURG 138 kV L is overloaded to 122.3% of its 218 MVA emergency rating for the loss of the LOWMOOR – HINTON AP 345 kV. The project contributes 14.5% DFAX (13.8 MW) to this overload.

For the loss of NEW HOPE AP – BARR LAKE 345 kV, the PARKE – TALOGA 138 kV L loads to 118.6% of its 331 MVA rating. Project DFAX: 9.4%.

PARKE 138 kV/1 kV Transformer Bank: 116.2% loading (338 MVA) following the outage of the BARR LAKE – NEW HOPE AP 345 kV; DFAX = 9.4%.

For the loss of BARR LAKE – KEAN MTN AND GRASSY CREEK 345 kV L, the STA 11 SUBSTATION – STOCK ISLAND 138 kV loads to 113.2% of its 242 MVA rating. Project DFAX: 22.3%.

For the loss of KEAN MTN AND GRASSY CRK – BARR LAKE 345 kV, the BATESVILLE – AFFINITY 138 kV loads to 112.3% of its 196 MVA rating. Project DFAX: 5.3%.

SPRINGBANK – STA 11 138 kV: 110.0% loading (234 MVA) following the outage of the KEAN MTN AND GRASSY CRK – BARR LAKE 345 kV; DFAX = 22.3%.

The POPLAR GAP – SANDY HOOK 138 kV L is overloaded to 108.4% of its 226 MVA emergency rating for the loss of the KEAN MTN AND GRASSY CRK – BARR LAKE 345 kV. The project contributes 5.3% DFAX (5.0 MW) to this overload.

MINNIX MTN – OPTICAL COATING LAB 138 kV: 108.3% loading (194 MVA) following the outage of the KEAN MTN AND GRASSY CRK – BARR LAKE 345 kV; DFAX = 8.1%.

For the loss of KEAN MTN AND GRASSY CREEK – BARR LAKE 345 kV, the ITMANN – WESMER 138 kV loads to 108.2% of its 234 MVA rating. Project DFAX: 5.3%.

For the loss of KEAN MTN AND GRASSY CREEK – BARR LAKE SUBSTATION 345 kV Line, the BAILEYSVILLE – ITANN 138 kV loads to 108.2% of its 236 MVA rating. Project DFAX: 5.3%.

For the loss of BARRLAKE – KEAN MTN AND GRASSY CRK 345 kV L, the SANDY HOOK – WESMER 138 kV Line loads to 107.7% of its 230 MVA rating. Project DFAX: 5.3%.

RECWAY – MINNIX MTN 138 kV: 105.7% loading (202 MVA) following the outage of the KEAN MTN AND GRASSY CREEK – BARR LAKE 345 kV; DFAX = 8.1%.

PRAIRIE CITY – RECWAY 138 kV: 104.3% loading (208 MVA) following the outage of the KEAN MTN AND GRASSY CREEK – BARR LAKE 345 kV line; DFAX = 8.1%.

The HUGHSON SUBSTATION – DIXLETA 345 kV is overloaded to 103.0% of its 1,350 MVA emergency rating for the loss of the RUSTIC – DIXLETA 345 kV. The project contributes 6.0% DFAX (5.7 MW) to this overload.

For the loss of NEW HOPE AP – MANATEE SUBSTATION 345 kV line, the PRINGLE – S. LAND PARK & 35TH AVE 138 kV loads to 102.8% of its 183 MVA rating. Project DFAX: 6.0%.

For the loss of KEAN MTN AND GRASSY CREEK – BARR LAKE 345 kV, the LONESOME PINE – MUNICIPAL LIGHT 138 kV loads to 102.3% of its 247 MVA rating. Project DFAX: 8.1%.

COAL – MCCLAIN ENERGY FACILITY 138 kV: 101.9% loading (217 MVA) following the outage of the MANATEE – HINTONAP 345 kV L; DFAX = 5.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
IOCO – KEAN MTN AND GRASSY CRK 138 kV	Rebuild line with higher-capacity conductor	\$7,337,000	3% (\$212,000)
HAYSI – HOBART 138 kV L	Rebuild line with higher-capacity conductor	\$10,263,000	2% (\$233,000)
KEAN MTN AND GRASSY CRK 345 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$18,948,000	2% (\$460,000)
KEAN MTN AND GRASSY CREEK 138/1kV TX	Replace transformer with higher-rated unit	\$5,764,000	2% (\$140,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NO TROY – LOST NATION 138 kV	Rebuild line with higher-capacity conductor	\$3,444,000	7% (\$233,000)
LYNN – LOST NATION 138 kV	Rebuild line with higher-capacity conductor	\$7,276,000	7% (\$491,000)
LYNN – ROGUE 138 kV	Rebuild line with higher-capacity conductor	\$8,462,000	7% (\$572,000)
ROGUE – TREEVANT 138 kV	Rebuild line with higher-capacity conductor	\$12,420,000	7% (\$839,000)
TREEVANT – SHULLSBURG 138 kV L	Rebuild line with higher-capacity conductor	\$2,232,000	7% (\$151,000)
TALOGA SUBSTATION – PARKE 138 kV	Rebuild line with higher-capacity conductor	\$8,034,000	4% (\$341,000)
PARKE SUBSTATION 138/1 kV Autotransformer	Replace transformer with higher-rated unit	\$3,071,000	4% (\$130,000)
STA 11 SUBSTATION – STOCK ISLAND 138 kV	Rebuild line with higher-capacity conductor	\$3,330,000	10% (\$345,000)
BATESVILLE – AFFINITY 138 kV	Rebuild line with higher-capacity conductor	\$24,543,000	2% (\$622,000)
SPRINGBANK – STA 11 138 kV	Reconductor line	\$4,473,000	10% (\$463,000)
POPLAR GAP – SANDY HOOK 138 kV L	Reconductor line	\$2,672,000	2% (\$68,000)
MINNIX MTN – OPTICAL COATING LAB 138 kV	Reconductor line	\$1,082,000	5% (\$58,000)
ITMANN – WESMER 138 kV L	Reconductor line	\$1,347,000	2% (\$34,000)
BAILEYSVILLE – ITANN 138 kV	Reconductor line	\$4,249,000	2% (\$108,000)
SANDY HOOK – WESMER 138 kV Line	Reconductor line	\$1,204,000	2% (\$31,000)
RECWAY – MINNIX MTN 138 kV	Reconductor line	\$1,866,000	5% (\$100,000)
PRAIRIE CITY – RECWAY 138 kV	Reconductor line	\$460,000	5% (\$25,000)
HUGHSON SUBSTATION – DIXLETA 345 kV	Reconductor line	\$17,302,000	6% (\$1,064,000)
PRINGLE – S. LAND PARK & 35TH AVE 138 kV	Reconductor line	\$1,717,000	5% (\$83,000)
LONESOME PINE – MUNICIPAL LIGHT 138 kV	Reconductor line	\$3,708,000	5% (\$198,000)
MCCLAIN ENERGY FACILITY SUBSTATION – COAL 138 kV	Reconductor line	\$3,457,000	5% (\$175,000)
	Total allocated to project		\$7,176,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.